Review of Transmission Lines – Part 3

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Lecture Objective

- By the end of this lecture, you will :
  - Recall concepts of semi-infinite transmission lines, load reflection coefficient, generalized reflection coefficient, and line impedance; and analyze transmission line problems using these concepts.

Semi-Infinite Transmission Line Connected to Source

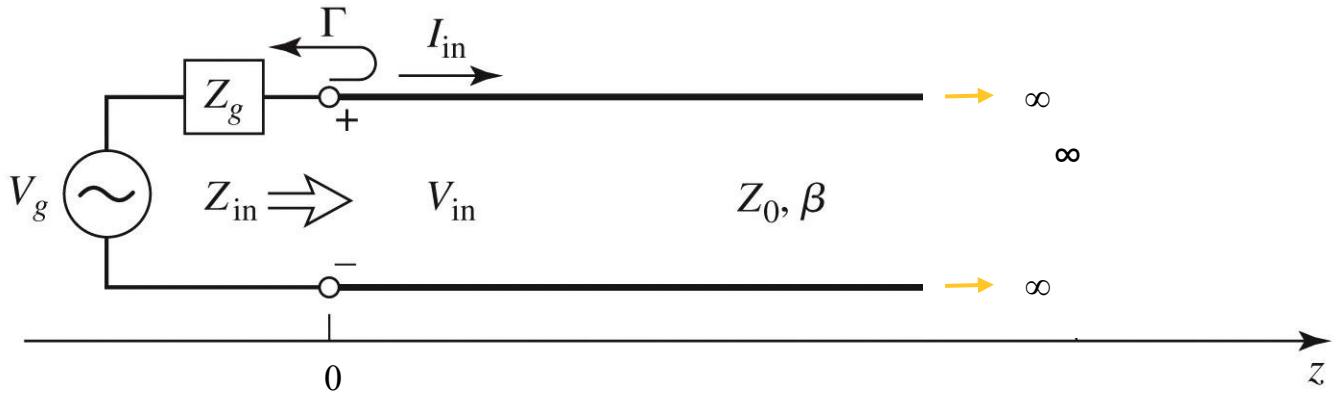

Figure 2.19
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(Adapted)

1. Semi-Infinite Transmission Line Connected to Source

- There is only a forward traveling wave on the line:

$$V(z) = V_0^+ e^{-\gamma z}$$

$$I(z) = \frac{V_0^+}{Z_0} e^{-\gamma z}$$

2. Semi-Infinite Transmission Line Connected to Source

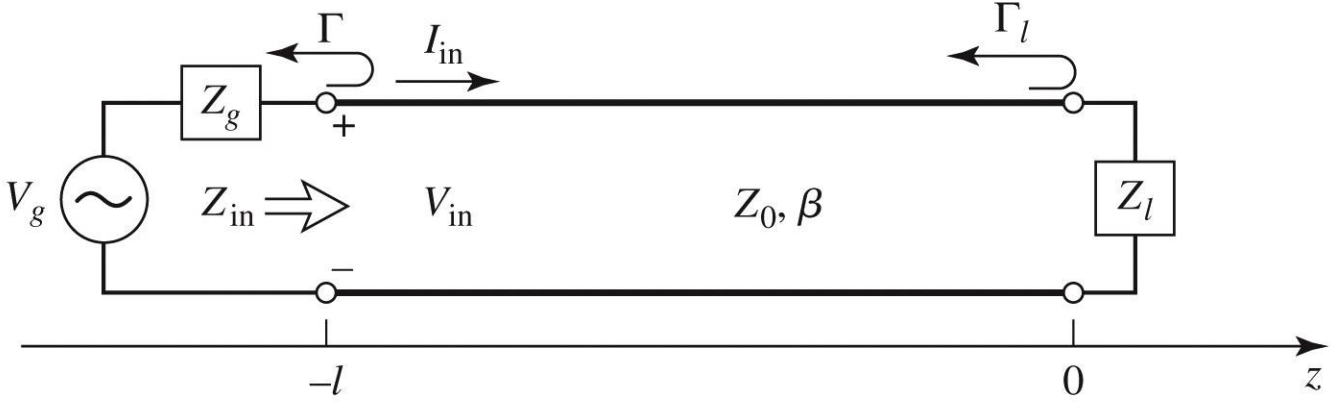
- The impedance looking into the semi-infinite line at $z = 0$:

$$Z_{in} = \frac{V(0)}{I(0)} = Z_0$$

- Hence

$$V_0^+ = \frac{Z_0}{Z_g + Z_0} V_g \qquad I_0^+ = \frac{V_g}{Z_g + Z_0}$$

Finite Transmission Line Connected to Source and Load



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2. Finite Transmission Line Connected to Source and Load

- There are (in general) both forward and backward traveling waves on the line:

$$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$

$$I(z) = \frac{1}{Z_0}\left(V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}\right)$$

3. Finite Transmission Line Connected to Source and Load

- Applying the BC at the load:

$$V(0) = V_0^+ + V_0^-$$

$$I(0) = \frac{1}{Z_0}\left(V_0^+ - V_0^-\right) = \frac{V(0)}{Z_L}$$

- Solving for the reflection coefficient at the load:

$$\Gamma_L = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Generalized Reflection Coefficient

- Define the generalized reflection coefficient:

$$\Gamma(z) = \frac{V^-(z)}{V^+(z)} = \frac{V_0^- e^{\gamma z}}{V_0^+ e^{-\gamma z}} = \Gamma_L e^{2\gamma z}$$

- In terms of generalized reflection coefficient, we have

$$V(z) = V_0^+ e^{-\gamma z}\left\{1 + \Gamma(z)\right\}$$

$$I(z) = \frac{V_0^+}{Z_0} e^{-\gamma z}\left\{1 - \Gamma(z)\right\}$$

2. Generalized Reflection Coefficient

- For a lossless line:

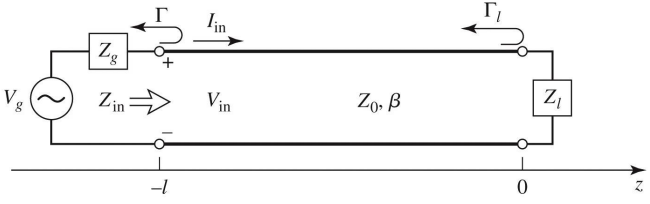
$$\gamma = j\beta$$

$$\Gamma(z) = \Gamma_L e^{j2\beta z}$$

3. Generalized Reflection Coefficient

- Example:

$$Z_0 = 50\,\Omega$$
$$Z_L = 25\,\Omega$$
$$\Gamma_L = \frac{25-50}{25+50} = -\frac{1}{3}$$
$$\Gamma(z) = \Gamma_L e^{j4\pi z/\lambda}$$



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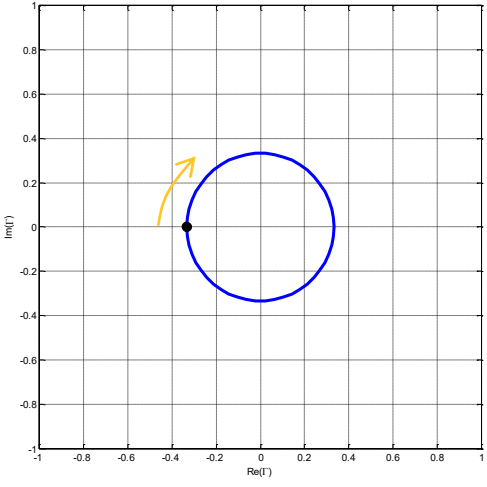


Line Impedance

- The impedance anywhere on the line is given by

$$Z(z) = \frac{V(z)}{I(z)} = Z_0 \frac{1+\Gamma(z)}{1-\Gamma(z)}$$

- The generalized reflection coefficient in terms of the line impedance is given by

$$\Gamma(z) = \frac{Z(z) - Z_0}{Z(z) + Z_0}$$

Standing Wave Ratio

- The voltage standing wave ratio on the line is given by

$$S = \frac{\left|V(z)\right|_{\max}}{\left|V(z)\right|_{\min}} = \frac{1+\left|\Gamma(z)\right|}{1-\left|\Gamma(z)\right|}$$

- For a lossless line

$$S = \frac{1+\left|\Gamma_L\right|}{1-\left|\Gamma_L\right|}$$

Example:

$Z_0 = 50\,\Omega$

$Z_L = 25\,\Omega$

$\Gamma_L = \frac{25-50}{25+50} = -\frac{1}{3}$

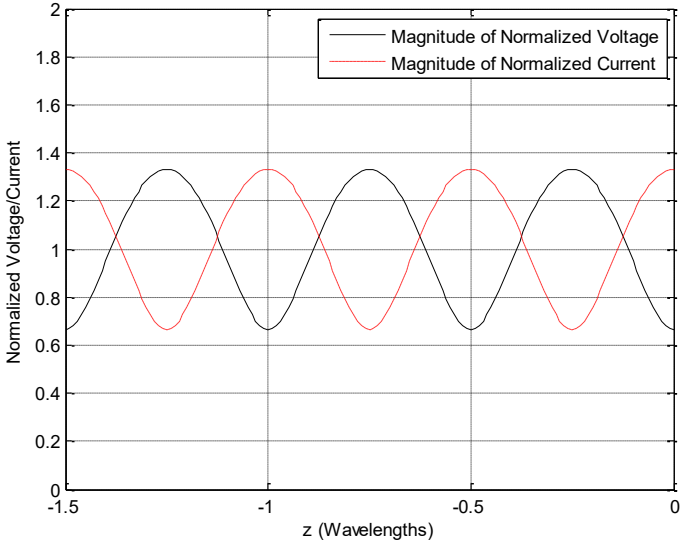
$S = \frac{4/3}{2/3} = 2 = \text{'2:1'}$

Standing Wave Patterns on Transmission Lines

Example:

$Z_0 = 50\,\Omega$

$Z_L = 25\,\Omega$



2. Standing Wave Patterns on Transmission Lines

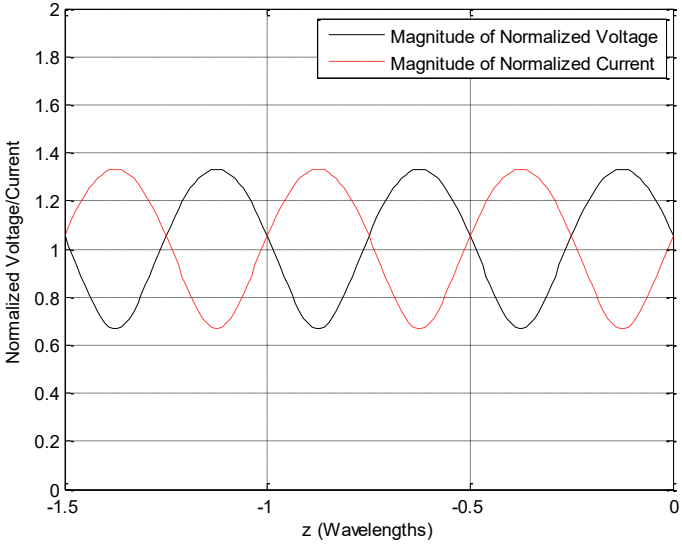
Example:

$Z_0 = 50\,\Omega$

$Z_L = 40 + j30\ \Omega$

$\Gamma_L = j\dfrac{1}{3}$

A complex load behaves just like a real load with some length of line added/subtracted.



Input Impedance of Transmission Line

$$Z_{in} = Z\left(z = -l\right) = Z_0 \frac{1 + \Gamma\left(-l\right)}{1 - \Gamma\left(-l\right)}$$

$$= Z_0 \frac{\left\{1 + \dfrac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}\right\}}{\left\{1 - \dfrac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}\right\}} = Z_0 \frac{\left\{\left(Z_L + Z_0\right)e^{\gamma l} + \left(Z_L - Z_0\right)e^{-\gamma l}\right\}}{\left\{\left(Z_L + Z_0\right)e^{\gamma l} - \left(Z_L - Z_0\right)e^{-\gamma l}\right\}}$$

2. Input Impedance of Transmission Line

$$Z_{in} = Z_0 \left\{ \frac{Z_L \cosh(\gamma l) + Z_0 \sinh(\gamma l)}{Z_L \sinh(\gamma l) + Z_0 \cosh(\gamma l)} \right\}$$

$$= Z_0 \left\{ \frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right\}$$

Special Case: Lossless Line

$$\gamma = j\beta \quad \Rightarrow \quad \tanh(\gamma l) = j\tan(\beta l)$$

$$\boxed{Z_{in} = Z_0 \left\{ \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} \right\}}$$

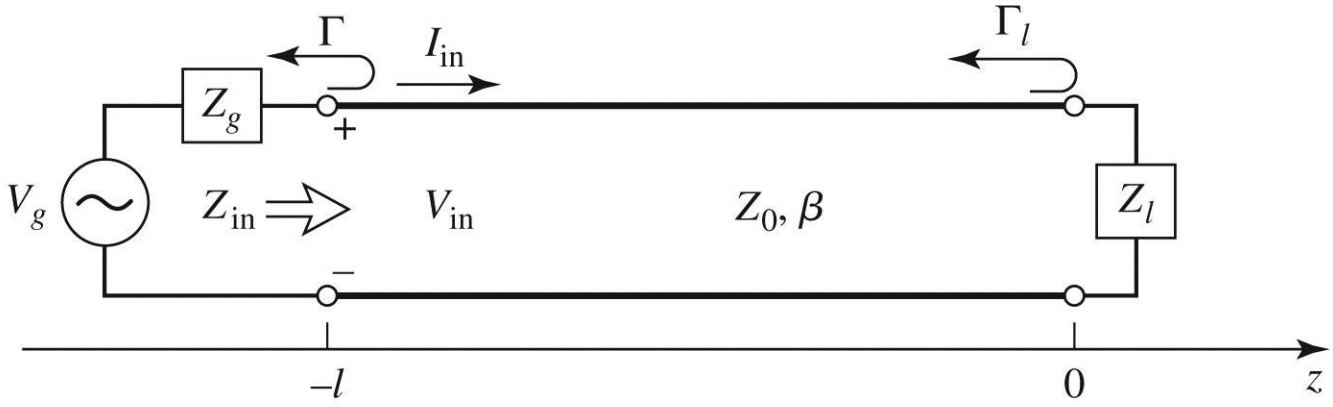
$$\beta l = \text{ electrical length of line}$$

Frequently expressed in degrees, e.g., an electrical length of 360 degrees corresponds to one wavelength.

Questions

- What is the equation for the input impedance of a lossless transmission line terminated in an open circuit?
- What is the equation for the input impedance of a lossless transmission line terminated in a short circuit?
- What is the equation for the input impedance of a lossless transmission line that is one-quarter wavelength long terminated in an arbitrary load impedance.

Transmission Line Circuits



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2. Transmission Line Circuits

- Using the BCs at both ends of the transmission line, we can write:

$$V(z) = \frac{Z_0 V_g}{Z_0 + Z_g} e^{-\gamma l} e^{-\gamma z} \left(\frac{1 + \Gamma_L e^{2\gamma z}}{1 - \Gamma_g \Gamma_L e^{-2\gamma l}} \right)$$

$$I(z) = \frac{V_g}{Z_0 + Z_g} e^{-\gamma l} e^{-\gamma z} \left(\frac{1 - \Gamma_L e^{2\gamma z}}{1 - \Gamma_g \Gamma_L e^{-2\gamma l}} \right)$$

$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}$$

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